

# Randy Zhu

Vancouver, BC | randy@randyzhu.com | github.com/RandoNandoz | linkedin.com/in/rzhu08 | Canadian Citizen · Open to Relocation

## EDUCATION

### University of British Columbia

April 2028

Bachelor of Science, Computer Science, Option in Software Engineering

GPA: 3.88/4

Teaching Assistant for Introduction to Computer Systems for 400+ students; Awarded Undergraduate TA Award

## EXPERIENCE

### Engineering Intern (Incoming)

August 2026 – December 2026

Nexthop AI

Bellevue, WA

### Firmware Developer Intern

May 2026 – August 2026

Netgear

Vancouver, BC

- Fixed screen tearing shipped in every product generation since 2012 by writing a new Linux fbdev driver for the ST7789S panel over Qualcomm's EB12 bus, synchronizing frame DMA to the panel's hardware TE signal
- Cut memory usage ~400MB and flash ~600MB by replacing a WebKitGTK device UI with a native LVGL/C implementation, building the navigation framework, widget library, and 20+ configuration screens
- Contributed key press events for every key upstream to LVGL, a widely used open-source embedded graphics library that previously supported only the enter key
- Enabled AI agents to develop and verify UI changes without target hardware by writing an SDL2 simulator backend with a mocked device-state layer and screenshot capture

### Research Intern

May 2025 – September 2025

University of British Columbia

Vancouver, BC

- Achieved a median speedup of 4.97x on large, open-source codebases with long-running integration tests by building a unit test generation tool targeting pytest using Python using dynamic execution tracing, AST transformations, and serialization
- Identified correct assertion generation, not execution capture, as the key barrier to automating carving on real test suites
- Integrated an LLM as a dependency classifier to auto-generate mocks for creating correct and deterministic unit tests

### Software Developer Intern

September 2024 – April 2025

Teck Resources

Vancouver, BC

- Eliminated ~100 hours of manual data entry by building a Dataverse-API ingest tool in C#
- Transformed a malformed SQL Server database with multivalued records stored as CSV strings in single columns into actionable business and safety insights, using T-SQL to parse the columns into separate columns and normalized tables
- Introduced the team to Playwright + NUnit to automate end-to-end testing, catching 87% of bugs before manual QA

## PROJECTS

### macOS iSCSI Driver | C++, DriverKit, IOKit, Swift

November 2025 – Present

- Building an open-source macOS iSCSI initiator that mounts remote storage as a local disk, providing a free, auditable alternative to \$249/license proprietary tools
- Bridging the Swift userspace networking client to the driver extension over a lock-free shared-memory ring buffer, saturating 2 10Gb SFP+ NICs, working around DriverKit's lack of network access
- Exposing the remote target as a local block device via the macOS IOUserSCSIParallelInterfaceController interface

### Racket Compiler | x86\_64 Assembly, LLVM IR, gdb, Racket

January 2026 – May 2026

- Cut compile times 180x (3h to 1m) by eliminating a pathological typed/untyped bottleneck in Typed Racket
- Replaced x86-only codegen with an LLVM IR backend, enabling compilation to any LLVM-supported platform
- Implemented closure conversion, tagged value representation, and tail call optimization via jump conversion

### Sasquatch | Swift, ARKit, LiDAR, Detectron2, Open3D, Python, FastAPI, PostgreSQL, GCP

March 2026 (24h Hackathon)

- Deployed an ML pipeline combining Detectron2 instance segmentation and LiDAR point clouds to detect climbing wall holds
- Implemented PCA-based wall orientation detection to project 3D point clouds to 2D for inference
- Reduced false positive rate from 90% to 1%** by adding a Gemini validation pass to filter non-hold wall objects

## SKILLS

**Languages:** C, C++, Python, Rust, TypeScript, Swift, Java, C#, SQL, Racket

**Technologies:** LVGL, React, Linux Kernel, OpenWRT, LLVM IR, DriverKit, IOKit

**Cloud/Infra:** AWS (EC2, ECS), GCP (Cloud Run, Cloud SQL, Gemini), Cloudflare (DNS, Pages), Docker, Linux

**Tools/Testing:** Git, GitHub Actions, CMake, GDB, pytest, JUnit